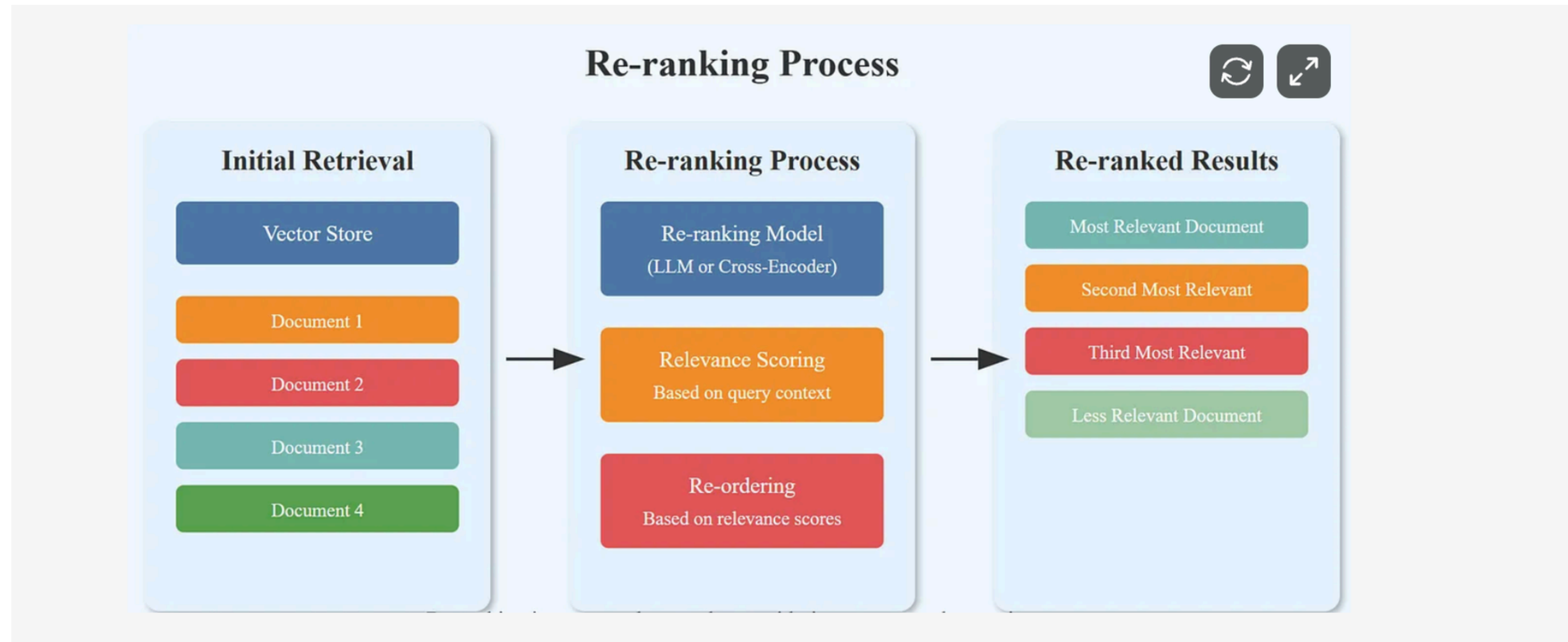


Week6

Reranker

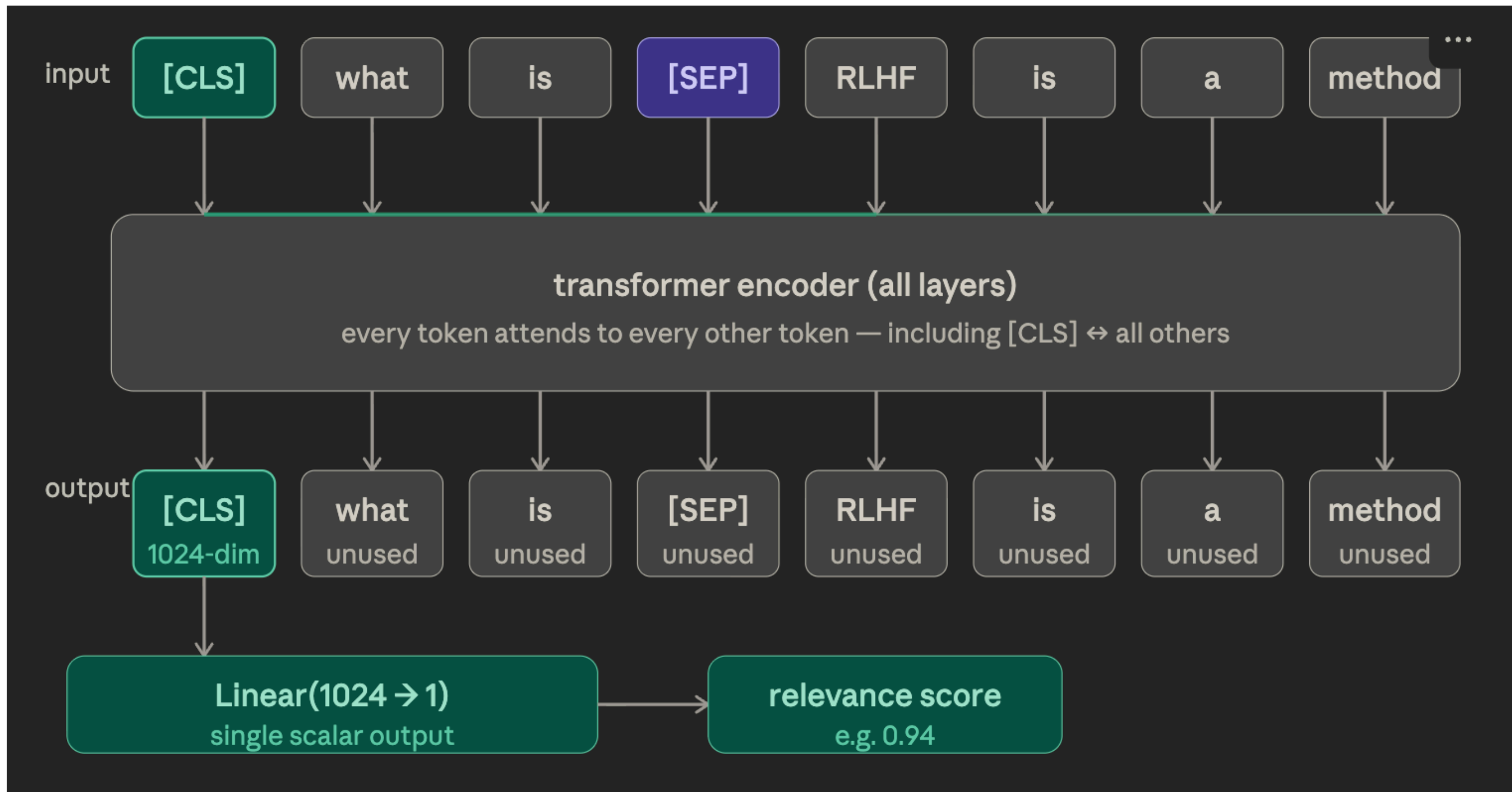


https://diamantai.substack.com/p/relevance-revolution-how-re-ranking?r=336pe4&utm_campaign=post&utm_medium=web&triedRedirect=true

- **Cohere (API)**
- **ColBERT**
- **BGE**

bge-reranker-base

Cross-encoder



mteb retrieval tasks

Smallest by queries:

- TRECCOVID: 171,332 docs – 50 queries
- TwitterHjerneRetrieval: 262 docs – 78 queries
- AILAStatutes: 82 docs – 50 queries (this is tiny on both sides)
- LegalBenchCorporateLobbying: 319 docs – 340 queries
- LEMBPasskeyRetrieval: 800 docs – 400 queries
- HagridRetrieval: 496 docs – 496 queries
- CovidRetrieval: 100,001 docs – 949 queries

Smallest by corpus size:

- AILAStatutes: 82 docs – 50 queries
- TwitterHjerneRetrieval: 262 docs – 78 queries
- LegalBenchCorporateLobbying: 319 docs – 340 queries
- HagridRetrieval: 496 docs – 496 queries
- LEMBPasskeyRetrieval: 800 docs – 400 queries

Without reranker

nDCG@1	0.9300
nDCG@3	0.9770
nDCG@5	0.9784

With Reranker
bge-reranker-base

nDCG@1	0.9050
nDCG@3	0.9588
nDCG@5	0.9652

k=10 rek=5

bge-reranker-large

nDCG@1	0.8850
nDCG@3	0.9624
nDCG@5	0.9640

bge-reranker-v2m3

nDCG@1	0.9500
nDCG@3	0.9835
nDCG@5	0.9835

max token count = 512, truncation

Token Length Report – BAAI/bge-reranker-large				
Total pairs analysed : 40000				
Pairs over 512 tokens: 47 (0.12%)				
	min	mean	max	>512
All pairs	42	198.3	541	47
Relevant pairs (s>0)	51	201.4	498	0
Distractor pairs (s=0)	42	197.8	541	47

nDCG@10 Scores Across Models

